

Practical No.: 6

Title: Data Extraction and Score Analysis from Web Table Using Selenium WebDriver

Aim

To automate the extraction of student marks from a dynamic web table using Selenium WebDriver and analyze the scores to determine:

1. Number of students scoring above 60 in at least one subject.
 2. Total number of students.
 3. Percentage of students meeting the criteria.
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Objectives

After completing this practical, students will be able to:

- Understand web table handling using Selenium WebDriver.
 - Extract data from HTML tables.
 - Perform data processing and analysis using Python.
 - Automate browser operations.
 - Generate analytical results from extracted web data.
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Software Requirements

Software	Version/Requirement
Operating System	Windows/Linux
Programming Language	Python 3.x
Automation Tool	Selenium WebDriver
Browser	Google Chrome
IDE	VS Code / PyCharm / IDLE

Theory

Selenium WebDriver

Selenium WebDriver is an open-source automation framework used to automate web browsers. It allows programs to interact with web elements such as:

- Text fields
- Buttons
- Links
- Tables
- Drop-down menus

Selenium communicates directly with the browser and performs user actions automatically.

Web Table

A web table is an HTML element used to display data in rows and columns.

HTML table structure:

```
<table>
```

```
<tr>
```

```
<th>Heading</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Data</td>
```

```
</tr>
```

```
</table>
```

Where:

- `<table>` → Defines the table.
 - `<tr>` → Defines a row.
 - `<th>` → Defines table heading.
 - `<td>` → Defines table data.
-

Data Extraction

Data extraction is the process of retrieving required information from a webpage.

In Selenium:

`find_elements()`

is used to extract multiple elements from a webpage.

Example:

```
rows = driver.find_elements(By.XPATH,"//table/tr")
```

Problem Statement

Create a student marks table containing:

Student Name Maths Science English

Amit	75	55	60
Priya	40	58	45
Rahul	62	70	65
Sneha	50	48	61
Rohan	30	45	50

Using Selenium:

1. Extract all student records.
 2. Check whether marks are greater than 60 in any subject.
 3. Count qualifying students.
 4. Calculate percentage.
-

Algorithm

Step 1:

Start the program.

Step 2:

Import Selenium libraries.

Step 3:

Launch Chrome browser.

Step 4:

Open the student marks web table.

Step 5:

Locate the HTML table using Selenium locators.

Step 6:

Extract student names and marks.

Step 7:

Check marks condition:

Maths > 60 OR

Science > 60 OR

English > 60

Step 8:

Increase counter if condition is satisfied.

Step 9:

Calculate percentage:

Step 10:

Display the result.

Step 11:

Close the browser.

Python Program

Data Extraction and Score Analysis from Web Table

Selenium WebDriver Python Practical

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import os
import time
print("Python script started")
```

```
# -----
```

```
# Step 1: Launch Chrome Browser
```

```
# -----
```

```
driver = webdriver.Chrome()
```

```
driver.maximize_window()
```

```
# -----
```

```
# Step 2: Open Student Marks HTML Page
```

```
# -----
```

```
file_path = os.path.abspath("student_marks.html")
```

```
print("Opening:", file_path)
```

```
driver.get("file://" + file_path)
```

```
time.sleep(2)
```

```
print("Web Page Loaded Successfully")
```

```
# -----
```

```
# Step 3: Extract Table Rows
```

```
# -----
```

```
# Get all rows from table
```

```
rows = driver.find_elements(  
    By.XPATH,  
    "//table[@id='marksTable']//tr"  
)
```

```
print("Total Rows Found:", len(rows))
```

```
# -----
```

```
# Step 4: Initialize Variables
```

```
# -----
```

```
total_students = 0
```

```
above_60_count = 0
```

```
print("\nStudent Marks Analysis")
```

```
print("-----")
```

```
# -----
```

```
# Step 5: Read Student Data
```

```
# -----
```

```
# Skip first row because it contains headings
```

```
for row in rows[1:]:
```

```
    columns = row.find_elements(By.TAG_NAME, "td")
```

```
    # Extract values
```

```
    name = columns[0].text
```

```
    maths = int(columns[1].text)
```

```
    science = int(columns[2].text)
```

```
    english = int(columns[3].text)
```

```
    total_students += 1
```

```
print(  
    name,  
    "Maths:",  
    maths,  
    "Science:",  
    science,  
    "English:",  
    english  
)
```

```
# Check if student scored above 60
```

```
# in at least one subject
```

```
if maths > 60 or science > 60 or english > 60:
```

```
    above_60_count += 1
```

```
# -----
```

```
# Step 6: Calculate Percentage
```

```
# -----
```

```
if total_students > 0:
```

```
    percentage = (above_60_count / total_students) * 100
```

else:

percentage = 0

-----

Step 7: Display Final Result

-----

print("-----")

print("Total Students:", total_students)

**print(
 "Students scoring above 60 in at least one subject:",
 above_60_count
)**

**print(
 "Percentage:",
 round(percentage, 2),
 "%"
)**

-----

Step 8: Close Browser

```
input("\nPress Enter to close browser...")
```

```
driver.quit()
```

Expected Output

Student Marks Analysis

```
>>>|===== RESTART: C:/Users/ditvi/OneDrive/Desktop/Software_Engineering_Bindiya/prac6
Python script started
Opening: C:\Users\ditvi\OneDrive\Desktop\Software_Engineering_Bindiya\student_marks.html
Web Page Loaded Successfully
Total Rows Found: 6

Student Marks Analysis
-----
Amit Maths: 75 Science: 55 English: 60
Priya Maths: 40 Science: 58 English: 45
Rahul Maths: 62 Science: 70 English: 65
Sneha Maths: 50 Science: 48 English: 61
Rohan Maths: 30 Science: 45 English: 50
-----
Total Students: 5
Students scoring above 60 in at least one subject: 3
Percentage: 60.0 %

Press Enter to close browser...
~::~|
```

Result Analysis

Parameter	Value
-----------	-------

Total Students	5
----------------	---

Students above 60	3
-------------------	---

Percentage	60%
------------	-----

Result

The Selenium WebDriver program successfully extracted student marks from a web table and performed score analysis. The program identified students scoring above 60 in at least one subject and calculated the percentage of students satisfying the condition.

Viva Questions

1. What is Selenium WebDriver?

Answer:

Selenium WebDriver is a browser automation tool used to automate web applications.

2. Which method is used to locate multiple elements?

Answer:

find_elements()

3. Why is XPath used?

Answer:

XPath is used to locate elements in an HTML document when other locators are not suitable.

4. What is the difference between find_element() and find_elements()?

find_element()	find_elements()
-----------------------	------------------------

Returns a single element	Returns a list of elements
---------------------------------	-----------------------------------

Used for one object	Used for multiple objects
----------------------------	----------------------------------

5. Why do we convert marks using int()?

Answer:

Web data is extracted as text. int() converts it into numbers for comparison and calculation.

6. How is percentage calculated?

Answer:

End of Practical

